

RE07 Global Infrastructure Investment and Management

Week 2

Institutional Frameworks and Stakeholder Governance

Professor Helen X. H. Bao

Department of Land Economy, University of Cambridge

Guest session: Department for Transport

What this week does

Week 1 ended with the finding that infrastructure projects systematically fail to deliver what was promised at the point of approval. This week asks what institutions exist to prevent that, and why they so often do not. The material divides between the machinery of public investment management, which is how a state chooses what to build, and economic regulation, which is how it controls what it has already built or handed over.

A	Why the state is in infrastructure at all	30 min
B	Public investment management: choosing what to build	40 min
C	Economic regulation and the price control	40 min
D	Delivery models and the PPP framework	40 min
E	Development institutions and the multilateral layer	25 min
F	Contracts that do not hold: renegotiation	30 min
G	Heathrow and the consenting of nationally significant infrastructure	35 min

What you already have

Where this week draws on modules you have taken, or are taking now. Nothing below is assumed: each idea is stated in full before the module is named.

Your programme	Where it was taught	What it gives this week
Real Estate Finance	RE01 session 3, capital structure and the weighted average cost of capital. RE02 Michaelmas lecture 5, key metrics and property indices.	A price control sets an allowed return on a regulatory asset base. That allowed return is a WACC, calculated by a regulator rather than by a company, which makes block C a finance problem rather than an administrative one.
Planning, Growth and Regeneration	PGR01 Michaelmas lectures 3 and 4, the UK planning system and the ideologies of planning law; lecture 2, national typologies. PGR04 lectures 1, 4 and 6, institutions, rent-seeking, and public goods.	The consenting regime behind Heathrow, the comparative method for placing one country's framework against another, and the account of why weak institutions produce bad projects.
Environmental Policy	EP02, environmental policy under asymmetric information, and elements of the theory of environmental policy I to III. EP07 week 4, adjudication systems.	Regulating an agent who knows more than you is the regulator's whole problem, instrument choice is the same question as delivery model choice, and a consent is final only once the courts have finished.
Everyone	RM01 lectures 7 and 8, qualitative methods and survey methodology.	Institutional quality is measured by index and expert survey. The PIMA framework and the Verhoest index are both built that way, and both inherit the weaknesses RM01 describes.

Block A

Why the state is in infrastructure at all

Four market failures, three institutional responses, and the question of which to use when.

Indicative length 30 minutes

The question the week has to answer

Infrastructure is provided almost nowhere by unregulated private markets. Even where assets are privately owned, prices are set by a regulator, entry is licensed, and the state retains a claim on service standards. That is unusual, and it is not explained by the assets being large or expensive: shipbuilding and semiconductor fabrication are both, and neither is regulated this way.

The explanation lies in four properties that recur across infrastructure sectors, each of which produces a specific market failure, and each of which has a standard institutional response. The rest of this block derives the institutions from the failures.

Four market failures, and what each implies

Natural monopoly

One firm can serve the market more cheaply than two, so competition in the market destroys value. Implies regulation of price and entry, or public ownership.

Externalities

Costs and benefits fall on parties outside the transaction, so private and social returns diverge. Implies subsidy, taxation, or a planning requirement.

Public good characteristics

Non-rival and hard to exclude from, so private supply is undersupply. Implies public funding, though not necessarily public provision.

Information asymmetry

The operator knows its own costs and the regulator does not. Implies incentive regulation rather than cost-plus, and is the reason regulation is hard rather than merely tedious.

What they have in common

Each is a reason for the state to be involved. None of them says how. Confusing the case for public funding with the case for public provision is the most common error in this field.

Natural monopoly is a claim about costs, not about size

A natural monopoly exists where the cost function is subadditive over the relevant range, meaning one firm can produce any given output more cheaply than any combination of firms. In network industries this arises because the fixed cost of the network is large and the marginal cost of an additional user is small.

Two consequences follow, and they pull in opposite directions. Competition in the market is wasteful, because duplicating the network raises total cost. But a single firm with no competitor and inelastic demand will price above cost. Regulation exists to resolve that tension, and every regulatory instrument in block C is an attempt to obtain the cost advantage without the pricing consequence.

Three responses to natural monopoly

Response	Mechanism	Weakness	Where it is used
Public ownership	The state owns and operates, and sets prices administratively	Weak cost discipline; investment competes with other public spending	Most road networks; rail infrastructure in many countries
Regulation of a private firm	Private ownership with a regulator setting allowed revenue and standards	Information asymmetry; regulatory capture; risk of underinvestment before a review	Water, energy networks and airports in England and Wales
Franchise bidding	Competition for the market rather than in it, by auctioning the right to serve	Contracts are incomplete, so the competition is only as good as the contract	Concessions, toll roads, most PPP arrangements

The bundling question

Hart

Incomplete contracts and public ownership: remarks, and an application to public-private partnerships, *The Economic Journal*, 2003.

Hart asks when building and operating should be bundled into one contract rather than procured separately. The answer turns on whether quality is contractible.

Bundle, and the builder internalises the operating consequences of its design choices, so it builds something cheaper to run. That is the case for the PPP. But the same incentive produces cost cutting wherever quality cannot be specified and enforced, and the operator captures the saving.

Bundling is therefore right where the service is measurable, as with a road surface or an available hospital bed, and wrong where it is not, as with the quality of care delivered inside that hospital.

Public private partnerships do not relax the budget constraint

The most common political justification for a PPP is that it allows investment without public borrowing. Engel, Fischer and Galetovic show that this is an accounting illusion. The state either pays the private partner over the life of the contract, in which case it has borrowed by another name, or it grants the right to charge users, in which case it has given away future revenue it could otherwise have collected.

What changes under a PPP is not the budget constraint but the allocation of risk and the incentives on delivery. If a PPP cannot be justified on those grounds, it cannot be justified at all.

What the state is actually buying

What a PPP can deliver

- Transfer of construction and completion risk to a party that can price and manage it
- Whole-life cost discipline, where design and operation sit with one party
- Delivery capability the public sector does not have in house
- A commitment device: a contract is harder to raid than a maintenance budget

What it cannot deliver

- Cheaper capital. The state borrows more cheaply than any consortium
- Escape from the fiscal consequences of the investment
- Transfer of risks the private party cannot control, such as demand or policy change, at a sensible price
- Good projects out of bad appraisal. A PPP delivers what was specified

Which failure bites, sector by sector

Sector	Natural monopoly	Externality	Public good	Dominant institutional response
Water and sewerage	Strong	Strong	Weak	Regulated private or public monopoly
Electricity networks	Strong	Moderate	Weak	Regulated private monopoly
Electricity generation	Absent	Strong	Absent	Competitive market with a carbon instrument
Roads	Strong locally	Strong	Strong for local roads	Public provision, tolling at the margin
Rail infrastructure	Strong	Strong	Weak	Public or regulated, with franchised operations
Fixed telecoms	Weakening	Moderate	Weak	Regulated access, with competition where feasible

Who exactly is the state

Speaking of the state as a single actor obscures the thing that matters. In any real project the public side is several bodies with different objectives: a sponsoring department that wants the asset, a finance ministry that is counting the liability, an economic regulator with a statutory duty to consumers, a consenting authority that must be procedurally fair, a local authority representing people who live nearby, and often a state-owned company that is a commercial party in its own right.

Coordination between these is itself an institution, and it is one of the fifteen assessed by PIMA. Where it fails, projects are approved that cannot be funded, or funded that cannot be consented.

Externalities have to be priced before they can be weighed

The case for public involvement often rests on effects that fall outside the transaction: emissions, noise, severance, agglomeration, and access for people who could not otherwise reach work. None of these has a market price, and all of them appear in the business case as numbers.

Where those numbers come from is a methodological question with a large literature behind it, covering revealed preference, stated preference and benefit transfer, and it is taught in EP02. For this module the important point is narrower: the externality valuations in an appraisal are estimates with wide confidence intervals, and they are frequently the component that determines whether a scheme passes.

Block A discussion

Seminar questions

1. Classify these services by whether quality is contractible, and therefore by whether bundling is appropriate: street lighting, a prison, a school building, teaching in that school, a water treatment works.
2. If the state borrows more cheaply than a consortium, under what conditions can a privately financed project still be the cheaper option overall?
3. Franchise bidding replaces competition in the market with competition for it. What has to be true about the contract for that substitution to work?

Block B

Public investment management

How a state decides what to build, and the institutions that stop it building the wrong thing.

Indicative length 40 minutes

Selection matters more than volume

Week 1 established that the aggregate return to infrastructure is real but modest, and that the distribution around it is wide. That combination has a direct implication: the value created by an investment programme depends far more on which projects are selected than on how much is spent.

Public investment management is the name for the institutions that do the selecting. It is unglamorous, it is where the returns are actually made or lost, and it is almost entirely absent from the finance literature on infrastructure.

The public investment management cycle

Stage	Question it answers	Typical failure
Planning	What do we need, and does this project belong to a strategy	Projects appear from political announcement rather than from a plan
Appraisal	Do the benefits exceed the costs, under honest assumptions	Optimism bias and strategic misrepresentation, as in week 1
Selection	Given a budget, which of the appraised projects proceed	Selection on political salience rather than on appraised value
Budgeting	Is the whole-life cost, including maintenance, provided for	Capital funded, maintenance not, so the asset degrades
Implementation	Is it being delivered to time, cost and specification	Weak monitoring, and no authority to stop a failing project
Evaluation	Did it deliver what was claimed, and what do we learn	Rarely done, and almost never fed back into appraisal

The framework this week uses

15

Institutions of public investment management assessed by the IMF's PIMA framework, across planning, allocation and implementation.

PIMA is a diagnostic instrument. It scores a country on the design of each institution and on its effectiveness in practice, and the gap between the two is often larger than the gap between countries.

Planning institutions cover fiscal rules, national and sectoral plans, coordination between levels of government, and the management of partnerships and of state company investment. Allocation institutions cover multi-year budgeting, budget comprehensiveness and unity, appraisal and selection. Implementation institutions cover funding availability, transparency, project management, asset monitoring and procurement.

Its teaching value is that it turns an abstract quality claim into a checklist a student can apply to a real country.

The efficiency gap

About 30%

Average gap between the public capital stock a country has and the stock its investment spending should have produced, on IMF estimates.

The efficiency gap is calculated by comparing countries on the relationship between cumulative investment spending and the value of the capital stock and services delivered. Countries close to the frontier convert spending into assets efficiently; others do not.

The order of magnitude matters more than the precise figure. If roughly a third of public investment spending does not become usable capital, then improving selection and delivery is worth more than any plausible increase in the budget.

The estimate is contested and depends heavily on how the capital stock is measured. It is a motivating figure, not a finding to lean on.

Norway builds the institution the evidence calls for

If cost underestimation is strategic rather than accidental, better forecasting will not fix it and an external check will. Norway operates exactly such a check. Projects above a threshold pass through two mandatory quality assurance stages before funding: the first interrogates the choice of concept, asking whether this is the right problem and the right kind of solution, and the second interrogates cost and management arrangements once a concept is chosen.

The reviews are carried out by external consultants under framework agreements with the Ministry of Finance, and they are conducted before political commitment rather than after. The first stage is the more important of the two, because a project that should not exist cannot be rescued by good cost control.

The United Kingdom's version

What exists

- The Green Book, which sets appraisal method across government, including optimism bias uplifts derived from reference class evidence
- The Infrastructure and Projects Authority, reviewing major projects and publishing delivery confidence assessments
- Gateway reviews at defined points in a project's life
- The Major Projects Portfolio, which makes the pipeline and its status visible

What is weaker than Norway

- Review generally follows the announcement of a project rather than preceding it
- The concept stage, asking whether this is the right kind of intervention at all, is less systematically challenged
- Delivery confidence assessments are advisory, and the authority to stop a project rests elsewhere
- Evaluation after completion is patchy, so the appraisal evidence base improves slowly

Why appraisal fails, and what each remedy actually fixes

- Technical error in forecasting is fixed by better models and better data. This is the least of the problem and the most commonly proposed remedy.
- Optimism bias, a psychological regularity, is fixed by reference class forecasting: adjust the estimate using the distribution of outcomes for similar projects rather than reasoning from the components of this one.
- Strategic misrepresentation is not fixed by either, because the promoter knows the true number and has an interest in not stating it. It requires an external check with the authority to stop the project, and accountability for the forecast after the fact.
- Weak institutions are fixed by none of the above, because the check itself is captured. This is the PIMA argument and the reason institutional quality, not analytical technique, is the binding constraint in many countries.

A pipeline is an institution, not a list

Investors say repeatedly that the constraint on infrastructure investment in developing markets is not capital but bankable projects. What they mean is that project preparation, which is expensive, slow and frequently wasted, is undersupplied.

A credible pipeline does three things at once. It tells investors that a flow of transactions exists, which justifies their building a local capability. It disciplines government by making commitments visible and comparable. And it separates preparation from procurement, so that projects reach the market already appraised. Where preparation is funded only after political commitment, appraisal becomes a justification exercise.

What a business case contains

Case	Question	Where it is usually weakest
Strategic	Is there a case for change, and does this fit the strategy	Written after the project was chosen, to justify it
Economic	Does this option maximise public value among the alternatives	The options appraised are variants of the preferred option
Commercial	Is the procurement deliverable, and will the market bid	Market appetite assumed rather than tested
Financial	Is it affordable, in capital and in whole-life terms	Capital affordability shown, operating and renewal cost not
Management	Can the organisation actually deliver it	Optimistic about capability, and rarely revisited

The discount rate does more work than any other assumption

An infrastructure appraisal compares costs incurred now with benefits accruing for sixty years. The rate at which those future benefits are discounted therefore determines the answer, and it is a value judgement expressed as a number.

The United Kingdom uses a social time preference rate built from pure time preference, expected consumption growth and the elasticity of marginal utility, and applies a declining schedule for very long horizons. Small changes matter enormously: at three and a half per cent a benefit sixty years out is worth about an eighth of its face value, and at one and a half per cent about two fifths. Any long-lived asset can be made to pass or fail by a defensible movement in this parameter.

Optimism bias uplifts, and where they come from

- United Kingdom appraisal guidance requires an explicit uplift to capital cost estimates at early stages, calibrated on the historic difference between estimated and outturn cost for comparable projects.
- This is reference class forecasting embedded in official procedure, and it is the institutional answer to the week 1 evidence.
- The uplifts are largest for the most novel and least standardised project types, which is where the evidence says they should be.
- The weakness is that an uplift applied to a deliberately understated base still produces an understated number, so the instrument corrects optimism and not misrepresentation.
- The other weakness is that uplifts are frequently negotiated down on the grounds that this project is different, which is what every promoter believes.

Evaluation is the stage nobody funds

Of the six stages, evaluation after completion is the least performed. The reasons are structural rather than accidental. The team that delivered the project has dispersed, the sponsoring minister has moved on, the counterfactual is hard to construct, and nobody has an interest in a document establishing that the benefits did not appear.

The cost of not doing it is that the appraisal evidence base never improves. Reference class forecasting requires a reference class, and a reference class requires that somebody recorded what happened. This is the strongest institutional argument for compulsory post-implementation review, and it is why the Flyvbjerg datasets exist at all: they were assembled by researchers because governments had not assembled them.

Block B discussion

Seminar questions

1. Norway reviews the choice of concept before political commitment. What would have to change in the United Kingdom for an equivalent check to bite, and who would resist it?
2. The efficiency gap suggests that a third of public investment does not become usable capital. If you had to improve one of the six stages of the cycle, which would you choose and why?
3. Project preparation is underfunded almost everywhere. Whose job is it to pay for it, and what does the answer imply about the role of development institutions?

Block C

Economic regulation and the price control

What a regulator sets, how it sets it, and why the answer is a weighted average cost of capital. Then an applied case on data centre demand for water and grid capacity.

Indicative length 65 minutes

The regulatory bargain

Where a network is privately owned and a natural monopoly, the state and the owner strike a bargain. The owner accepts that its prices, service standards and investment obligations are set by an independent regulator. In return the regulator commits to allowing revenue sufficient to finance the efficient operation of the network and to earn a return on the capital invested in it.

The bargain is credible only if it survives political pressure, which is why regulators are made independent, and only if the owner believes the return will still be there in fifteen years, which is why regulatory precedent matters more than regulatory rules.

The building blocks of an allowed revenue

Component	What it covers	Where the argument is
Return on the regulatory asset base	The cost of the capital tied up in the network	The weighted average cost of capital, and what counts as being in the asset base
Return of capital, or depreciation	Recovery of the investment over the asset's life	Asset lives, which determine how quickly customers pay for an asset
Operating expenditure allowance	Efficient running costs for the period	What is efficient, benchmarked against comparable firms
Incentive adjustments	Rewards and penalties against output targets	Whether the targets were demanding, and whether outperformance is skill or forecast error

The allowed return is a weighted average cost of capital



The single most consequential number in a price control is the allowed return, and it is a weighted average cost of capital calculated by a regulator rather than by a company. Gearing, the cost of debt, the risk-free rate, the equity risk premium and the beta all appear, and each is contested at every review, because a difference of fifty basis points on a large asset base is worth a great deal of money.

For students who have taken RE01 this is a familiar calculation in an unfamiliar setting. The unfamiliar part is that the number is not discovered in a market: it is determined administratively, published, and then defended in front of an appeal body.

Two regulatory regimes

Rate of return regulation

- The firm recovers its costs plus an allowed return on its asset base
- Low risk for the firm, and low incentive to reduce cost
- The Averch and Johnson result: a regulated firm over-invests in capital relative to the efficient mix, because the return is earned on the asset base
- Historically dominant in the United States

Incentive regulation

- Prices are capped for a period, and the firm keeps what it saves within it
- Strong cost incentive within a period, and a ratchet problem across periods
- Requires the regulator to forecast efficient cost, which is where the information asymmetry bites
- The RPI minus X framework, developed in the United Kingdom and now widespread

The regulator's problem in one sentence

- The regulator must set an allowance for costs it cannot observe, on the basis of information supplied by the party whose payment depends on that information.
- The standard responses are benchmarking against comparable firms, menus that make it profitable to reveal a true cost forecast, and outcome-based targets that avoid specifying inputs at all.
- Each response is a mechanism design solution to an adverse selection problem, which is the EP02 material applied to a network utility rather than to a polluter.
- None of them removes the asymmetry. They shift how much of the resulting rent the firm retains.

What can go wrong with a regulator

Capture

The regulator comes to share the industry's view of what is reasonable, through information dependence and career flows in both directions.

Political interference

Prices are salient. A government facing an election has an interest in the outcome of a price review and often has instruments to influence it.

Hold-up in reverse

Once the capital is sunk, the state can allow a lower return than was implied when the investment was made. Rational investors anticipate this and demand more.

Ratchet effects

A firm that reveals what efficiency is achievable is given a harder target next time, so it does not reveal it.

Complexity

A regime with enough special provisions becomes unintelligible to anyone outside it, which weakens accountability and favours incumbents.

Underinvestment

Where the allowed return is set below the true cost of capital, the firm invests less than the regime intends and quality falls with a long lag.

A case worth carrying through the module

The water and sewerage industry in England and Wales is the most studied regulated network in the world, and every element of this block is visible in it: a privatised natural monopoly, a regulatory asset base, five-yearly price reviews, an allowed return that is publicly argued over, comparative benchmarking across companies, incentive mechanisms on leakage and pollution, an appeal route to the competition authority, and a political environment that has become hostile.

It is also a case where the financial structures adopted by owners, including high gearing and complex holding company arrangements, have become the subject of the regulatory argument rather than a private matter. That makes it the right case for a course about investment as well as regulation.

The regulatory asset base is the thing investors actually buy



A regulated network company is, financially, a claim on a regulatory asset base. Capital expenditure approved by the regulator is added to the base, depreciation removes it, and inflation indexation raises it. The company earns the allowed return on whatever the base is, so the investment case reduces to two questions: how fast will the base grow, and what return will be allowed on it.

This has a consequence students find counter-intuitive. A regulated utility is rewarded for investing, because investment enlarges the base on which it earns. Where the regulator is a weak scrutineer of capital expenditure, that is the Averch and Johnson distortion in operation, and it is why capital plans are examined so much more aggressively than operating costs.

How modern price controls have changed



- Total expenditure rather than separate capital and operating allowances, which removes the incentive to classify spending as capital in order to earn a return on it.
- Outcome-based targets specified as what customers experience, rather than as inputs the company must buy.
- Menu regulation, in which a company chooses from a schedule of forecast and incentive combinations, and the choice reveals its private cost estimate.
- Comparative benchmarking across companies, which substitutes for the information the regulator does not have.
- Longer control periods, which strengthen the within-period incentive and worsen the forecasting problem.

Airport regulation, and why it matters for the guest session



A designated United Kingdom airport is regulated by the Civil Aviation Authority, which sets a cap on charges to airlines using the same building blocks as any other price control: an asset base, an allowed return, forecast passenger volumes and efficiency targets. The regulated entity is privately owned and its investment case depends on the determination.

The relevance to Heathrow expansion is direct. Consent and regulation are separate processes on separate timetables with separate decision-makers, and a promoter needs both. A consent without a determination that permits recovery of the cost is worth nothing, and a determination cannot sensibly be made before the scheme is defined. Managing that sequencing is one of the hardest practical problems in the case.

An applied case, and what it is a case of

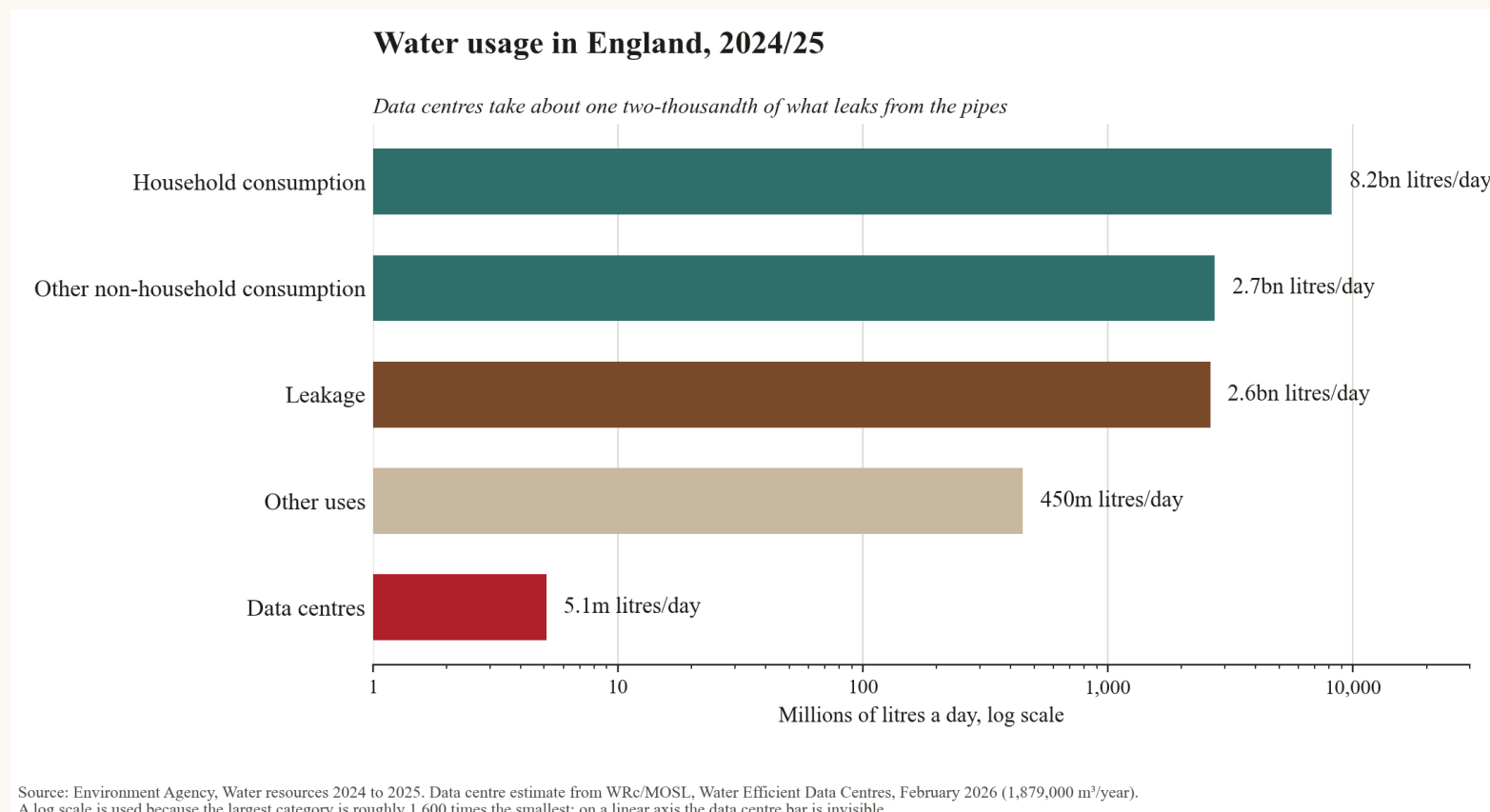
The rest of this block works a single live example: the demand that data centres place on water and on grid capacity. It is placed here because the question it ends on, which is who pays when one user requires the network to be reinforced, is a price control question and belongs with the material above.

The case is not really about data centres. It is about a measurement problem that recurs across every sector in this module. A national aggregate can be honestly compiled, correctly reported, and still answer a question nobody was asking, because infrastructure demand does not arrive as a share of a national total. It arrives as a point load on one local system.

Water use in England by category, 2024/25



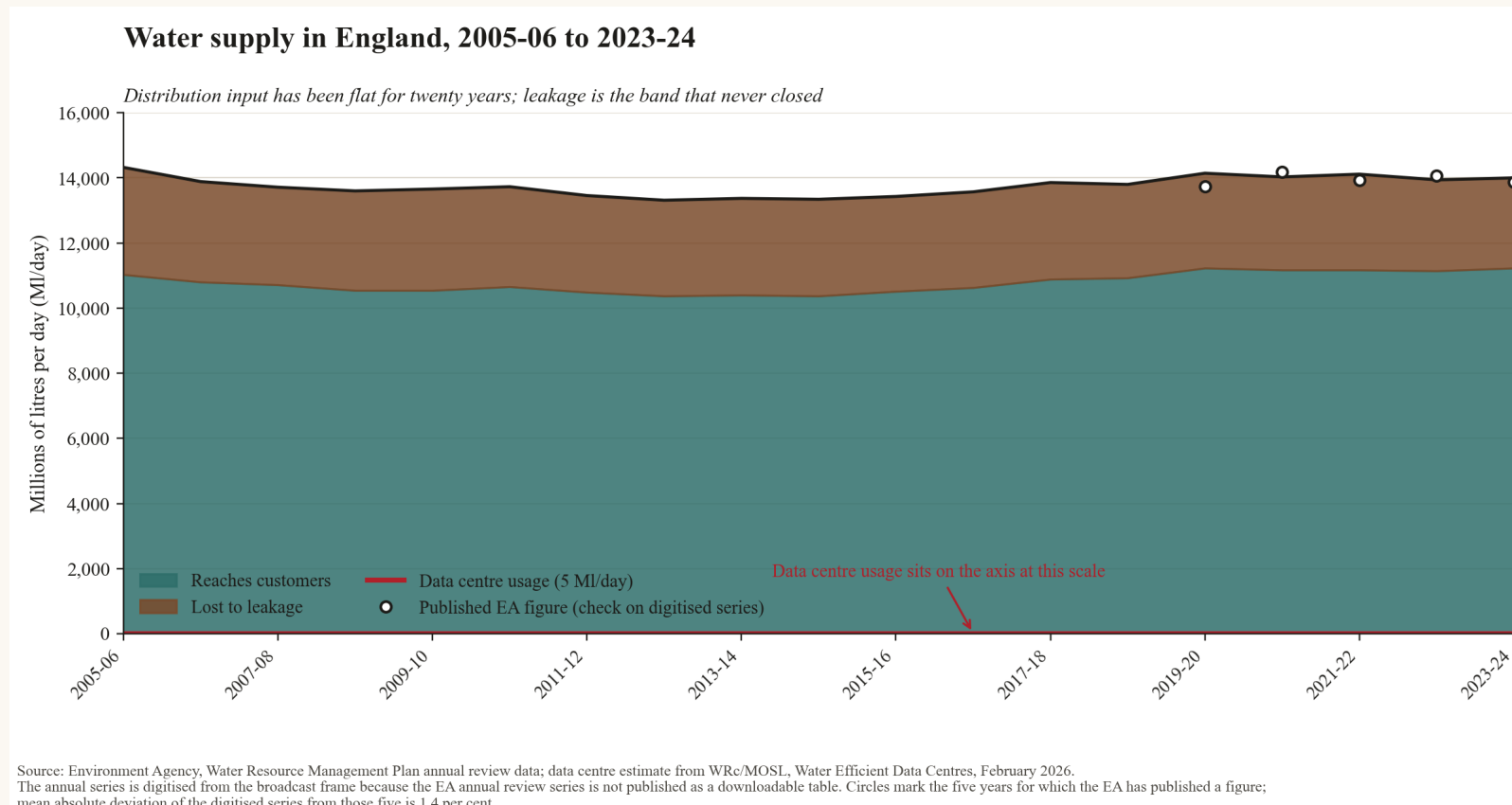
Public argument about data centres has focused on water. This is the whole English picture for 2024/25, on a logarithmic scale because on a linear one the data centre bar cannot be seen at all.



Distribution input and leakage in England, 2005 to 2024



The same conclusion over twenty years. Total water put into supply has been flat, leakage has barely moved, and data centre use is too small to form a visible band.



Environment Agency, Water Resource Management Plan annual review data.

What question do those two charts answer?



Seminar questions

1. State the question the charts answer precisely, in one sentence, before deciding whether it is the right question.
2. A campaigner objects to a specific data centre on water grounds. Do these charts refute that objection?
3. What would you need to measure instead, and at what geography, for the objection to be tested rather than dismissed?

Data centre locations against water stress classification

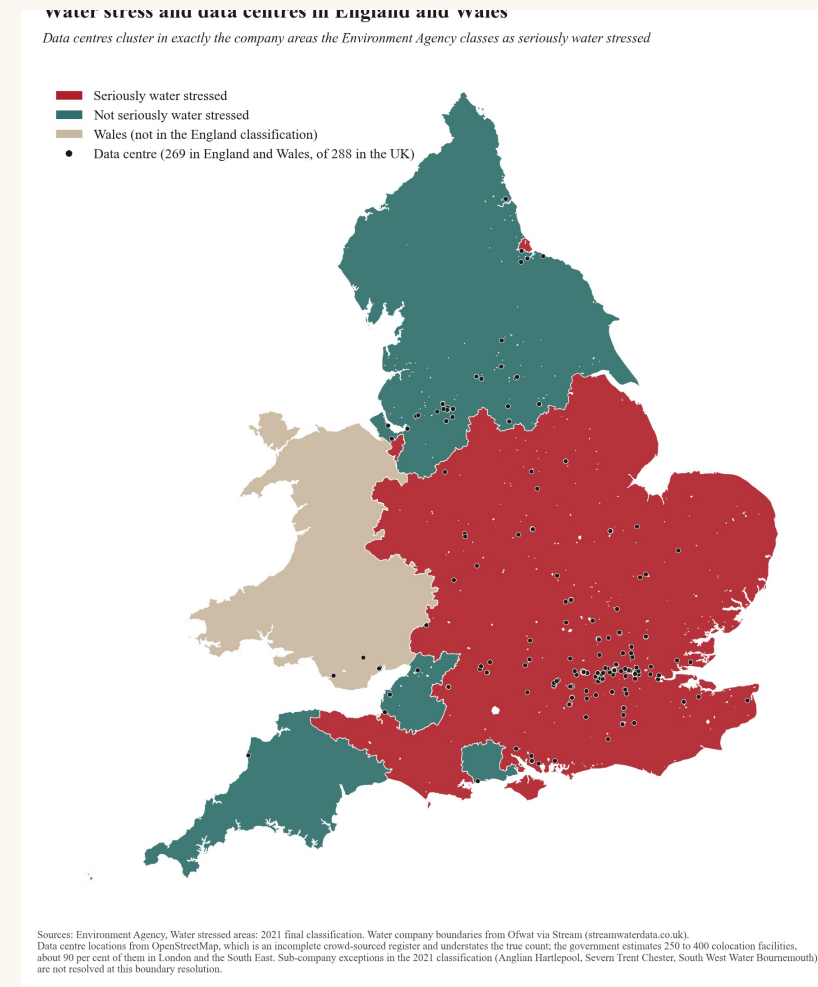


The same evidence, disaggregated.

Areas shaded dark are those the Environment Agency classified in 2021 as seriously water stressed. The points are the data centres recorded in OpenStreetMap.

Nothing here contradicts a number on the previous two slides. The national total really is small. It is simply that the national total was never the quantity that determined whether any particular scheme could be built.

What a water company faces is not a share of a national aggregate but a specific abstraction request, in a specific catchment, in a specific dry year.



Environment Agency water stressed areas classification, 2021; facility locations from OpenStreetMap.

The concentration the aggregate concealed

83 per cent

of data centres in England and Wales sit in seriously water stressed areas

Of the 269 facilities OpenStreetMap records in England and Wales, 224 fall inside water company areas the Environment Agency classified as seriously water stressed in 2021. Against the full United Kingdom count of 288 the share is 78 per cent. The classification does not extend to Scotland or Northern Ireland, which is why the smaller denominator is the meaningful one, and why it should be stated whenever the figure is used.

Within the sector, the same problem again



Measure	Value	What it implies
English data centres using under 1,000 m ³ a year	67 per cent, mean 217 m ³	Most facilities use about as much water as a 600 m ² office
Share of sector water use taken by the largest six	65 per cent	Sector water use is a handful of sites, not a distribution around a mean
Share taken by the largest fifteen	over 80 per cent	A sector average is close to meaningless as a planning input
Of those six largest, in service since 2020	five, two of them in 2024	The concentration is recent and still moving

A quantity that is metered, billed, and still disputed



Five-fold

the gap between two credible published estimates of the same figure

WRc and MOSL put English data centre water use at 1,879,000 m³ a year. Foxglove and The Times reported water companies supplying almost 10 billion litres a year to 231 facilities. The quantity is metered at the point of supply and invoiced monthly, so the discrepancy is not a measurement difficulty. Nobody is obliged to report it, which is why the report's central recommendation is a mandatory reporting framework.

The grid position, which is the binding constraint

Quantity	Figure	What it is, and is not
Data centre projects seeking connection, per Ofgem	around 140, roughly 50 GW	Projects that have applied, not projects that will be built
All large-user applications, end July	125 GW, up from 41 GW under a year earlier	An application register, tripling in under a year
Data centre share of that total	at least 80 GW	Larger than the whole system at peak
British peak electricity demand, 2023	58 GW	The system the queue is queuing for

Non-domestic electricity demand per square kilometre

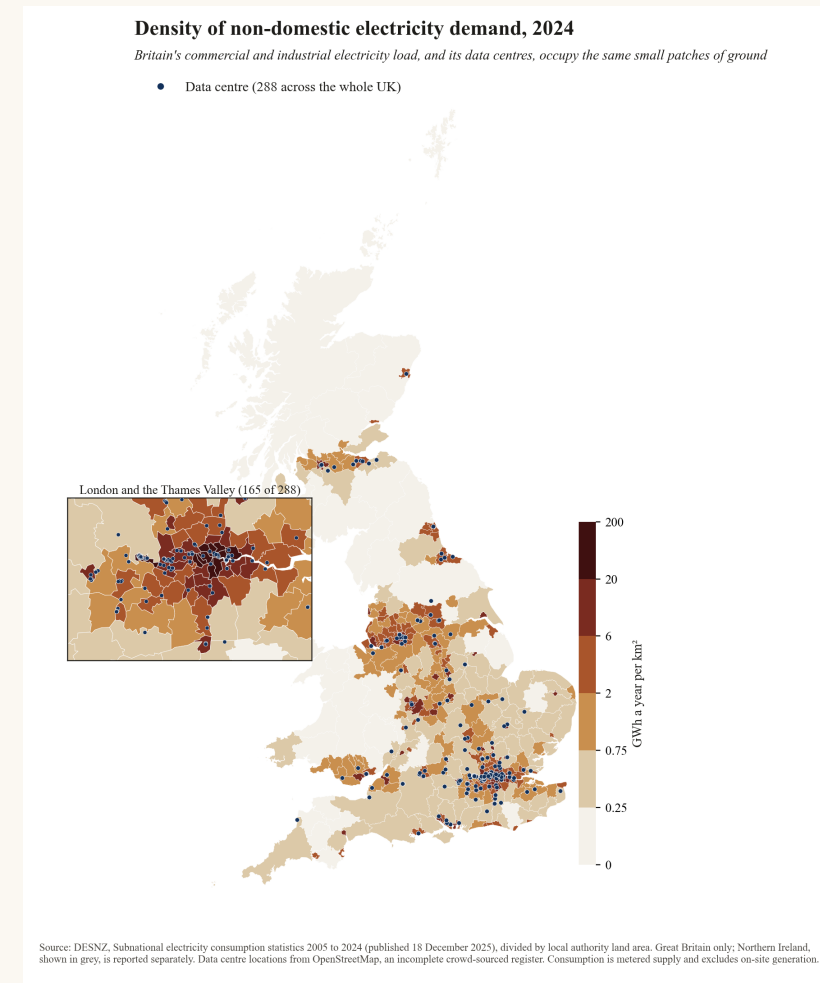


The same concentration, seen from the electricity side.

Density rather than total, because density is what a distribution network actually has to carry. A large authority with moderate demand poses no problem; a small one with concentrated demand does.

The top thirty of 350 British local authorities take 26.5 per cent of all non-domestic consumption. Slough, singled out in the case, draws 1,786 GWh across about 32 square kilometres.

Fifty-seven per cent of the mapped facilities fall inside the inset alone.



DESNZ sub-national electricity consumption, 2024. Inset covers London and the Thames Valley.

A connection queue is a free option on scarce capacity

Seminar questions

1. If applying to connect is cheap and refundable, what will the size of the queue tell you about real demand? Why is over-subscription the expected outcome rather than a surprise?
2. Three ways to ration scarce capacity: price the option, allocate by queue position, or allocate by readiness to build. Take one each and argue it, including who it disadvantages.
3. A single hyperscale user requires network reinforcement. Who should pay, the user or the regulated asset base and therefore every bill payer? Does the answer change if the facility is designated critical national infrastructure?
4. American operators have signed a pledge to fund their own connection infrastructure rather than socialise it. Would you expect that commitment to survive a downturn, and what would make it credible?

Block C discussion

Seminar questions

1. A regulator lowers the allowed return by fifty basis points. Trace the consequences for the company's share price, its investment programme, and its customers over the following ten years.
2. Incentive regulation gives strong cost discipline within a period and creates a ratchet across periods. How would you design a regime that mitigates the ratchet without abandoning the incentive?
3. Is the risk that a regulated network faces genuinely lower than that of an unregulated firm, or has it merely been converted from market risk into political risk?

Block D

Delivery models and the PPP framework

What the state is contracting for, the cycle it runs to get there, and the framework that has to exist first.

Indicative length 40 minutes

The delivery spectrum, with what transfers

Model	Who designs and builds	Who operates	Who bears demand risk	Who owns at the end
Traditional procurement	Contractor, to the state's design	The state	The state	The state throughout
Design and build	Contractor, to an output specification	The state	The state	The state throughout
Availability PPP	Consortium	Consortium	The state	Transfers to the state on expiry
Concession	Consortium	Consortium	The concessionaire	Transfers to the state on expiry
Privatisation with regulation	The owner	The owner	Shared through the price control	The owner, indefinitely

The label is not the transaction

The vocabulary in this field is unreliable. The same structure is called a PPP in one country, a concession in another and a private finance initiative in a third, and identical labels describe materially different risk allocations. Delmon's advice is to ignore the label and read the risk allocation, and it is sound.

Four questions identify any structure. Who is paid, and by whom. What triggers the payment: use of the asset, or its availability. What happens if the asset is not available. And what happens at expiry. Answer those and the label is irrelevant.

The PPP project cycle

Stage	What happens	Where it goes wrong
Selection and prioritisation	The project is identified, screened against a plan, and tested for PPP suitability	Projects selected because they can be financed off balance sheet rather than because they should be
Development	Feasibility, business case, market sounding, structuring, advisers appointed	Preparation underfunded, so the project reaches the market half formed
Procurement and award	Prequalification, bidding, evaluation, negotiation, financial close	Too few bidders; changes during negotiation that undo the competition
Implementation	Construction, commissioning, operation, contract management, handback	Contract management resourced far below what the contract requires

What a PPP framework has to contain

Legal framework

- Authority for the contracting agency to enter the transaction at all
- Ability to give the assurances lenders require, including on termination compensation
- Procurement rules that permit the process the transaction needs
- Land acquisition powers, and a route to secure title
- Enforceable security over assets and revenues, and a workable dispute resolution route

Institutional framework

- A unit with responsibility for the programme and the capacity to run it
- Approval gateways with genuine authority to stop a transaction
- Fiscal risk management, so contingent liabilities are recorded and priced
- A project team that survives the transition from procurement to operation
- Monitoring and evaluation that feeds back into the next transaction

PPP units and where they sit

Most countries running a PPP programme create a dedicated unit. Where it sits determines what it can do. A unit inside the finance ministry can enforce fiscal discipline and refuse transactions, but is often distrusted by line ministries and can become a bottleneck. A unit inside a line ministry has sector knowledge and a promotional interest, which is exactly the conflict the gateway is supposed to prevent. A unit outside government as an agency has independence and usually lacks authority.

The design question is whether the unit is a promoter or a gatekeeper. Asking it to be both, which is common, produces a body that approves its own transactions.

Fiscal risk is the part that is usually missed

A PPP creates future obligations for the state: availability payments, minimum revenue guarantees, termination compensation, and exposure to events specified in the contract. Where these are not recorded, priced and reported alongside conventional debt, a government can accumulate a liability it does not know the size of.

The consequence is not hypothetical. Several countries have discovered guarantee exposures that materially changed their fiscal position, and the discovery generally coincided with the event that triggered them. Recording contingent liabilities is therefore not an accounting nicety but the mechanism that stops the off-balance-sheet motive from driving project selection.

The evaluation evidence is contradictory

Contested

Hodge and Greve's verdict after reviewing the international performance evidence on public private partnerships.

The honest position on PPP performance is that the evidence does not settle it. Studies differ on the counterfactual, on whether transaction costs are counted, on whether the comparison is with the public sector as it actually performs or as it is assumed to perform, and on whose account of risk transfer is accepted.

Hodge and Greve add a governance argument that is independent of the value for money question. Long-term contracts move decisions out of the political arena for decades, which is a benefit if you think politics is the problem and a cost if you think accountability matters.

Students should leave able to state the case on both sides rather than holding a position they cannot defend.

Measuring how supportive a framework is

Verhoest and colleagues build a governmental support index across twenty European countries, scoring the extent to which the national institutional framework is conducive to public private partnerships. The instrument covers policy and political commitment, the legal and regulatory environment, the presence and role of dedicated units, and practical support for procuring authorities.

For a comparative exercise this is more useful than a narrative, because it forces the analyst to say what an institutional framework consists of and to score each element. The planning students will recognise the method from the typologies of national planning systems in PGR01.

Value for money and the public sector comparator

The standard test asks whether a privately financed option delivers better value than conventional procurement. It is operationalised by building a public sector comparator: a costing of the same output delivered publicly, adjusted for the risks the public sector would retain.

The method is defensible in principle and fragile in practice, because the answer is determined by the risk adjustment, and the risk adjustment is an assumption. Where the comparator is constructed after a policy preference has been formed, it tends to confirm it. The most useful discipline is to require that the risk adjustment be stated as a probability distribution over specific events rather than as a single percentage.

Unsolicited proposals

A private party approaches government with a project nobody asked for. The attraction is genuine: the proponent may have identified something the state had not, and may bring a technology or a site nobody else has.

The difficulty is that there is no competition, no comparison and no market price, and the state is negotiating with the only party that has done the work. The usual responses are to run a competitive process after the fact with a bonus or right to match for the originator, or to buy the intellectual property and tender the project properly. Both are imperfect, and countries that have accepted unsolicited proposals at scale have generally regretted it.

Contract management after financial close

- A twenty-five year contract requires an active counterparty for twenty-five years. In practice the public side is usually resourced for the procurement and not for the operation.
- The transition itself is the risk point: the team that negotiated the contract disperses, and the people who inherit it did not write it and often cannot find the schedules.
- Payment mechanisms have to be operated, deductions applied, and performance monitored. A deduction never applied is a term that does not exist.
- Changes are inevitable over a long contract, and each one is a bilateral negotiation with a party that knows the state cannot walk away.
- Handback at expiry, with the asset in a specified condition, is planned for far too late where it is planned for at all.

Block D discussion

Seminar questions

1. A minister wants a hospital delivered as a PPP because the capital cost will not score against the borrowing target. What should the finance ministry say, and what institution should have prevented the question from arising?
2. Should a PPP unit be a promoter or a gatekeeper? If it cannot be both, which function should sit with the unit and where should the other go?
3. The performance evidence on PPPs is contradictory. What study design would actually settle the question, and why has nobody run it?

Block E

Development institutions and the multilateral layer

Who lends where commercial capital will not, and what they supply beyond money.

Indicative length 25 minutes

The institutional layer

Multilateral development banks

Owned by member states, lending to sovereigns and to the private sector. The World Bank Group, the regional banks, and newer entrants such as the AIIB and the New Development Bank.

Development finance institutions

Investing in private projects with a development mandate. The IFC, and bilateral institutions such as British International Investment and DEG.

Export credit agencies

Supporting exports from their home country through direct lending, guarantees and insurance. Often the only source of long tenor debt in difficult jurisdictions.

Bilateral aid agencies

Grant and concessional funding, frequently for project preparation, which is the stage the market does not fund.

Climate funds

Concessional capital targeted at mitigation and adaptation, with their own eligibility rules and appraisal processes.

What they share

A mandate that is not a required return, which explains behaviour that looks irrational to a commercial investor and is not.

What they supply beyond capital

- Tenor. Commercial lenders will not lend for twenty years in most emerging markets; a development institution will, and its participation lengthens what others will offer.
- Political risk absorption. Their presence in a syndicate changes the calculus of expropriation and of contract breach, which is the empirical finding in Hainz and Kleimeier.
- Preferred creditor treatment. Sovereigns have historically serviced multilateral debt through crises, which lowers expected loss for everyone in the structure.
- Standards. Environmental and social requirements propagate through the syndicate and into the project documents, which is regulation by lending.
- Preparation. Feasibility work, transaction advice and capacity building, which is undersupplied because it is expensive and often wasted.

The mobilisation claim

7 to 1

Bank credit crowded in per dollar of multilateral lending over a three-year period, on syndicated loan data for developing countries between 1993 and 2017.

Broccolini and colleagues estimate mobilisation effects directly rather than by assertion. Multilateral participation raises the size of bank inflows, the number of lenders, and the average maturity of syndicated loans, and the effects persist for up to three years without being offset by a decline in bond financing.

The result matters because the entire blended finance agenda rests on the claim that concessional capital brings commercial capital with it. Until this paper the claim was largely institutional self-reporting.

It is also a ceiling on expectations. Seven to one is a good multiplier and it is not the hundred-to-one implied by some official rhetoric about closing the investment gap.

Preferred creditor treatment, and what sustains it

Sovereign borrowers have historically continued to service multilateral debt through crises in which they defaulted on commercial obligations. Nothing in law requires this. It is sustained because a country that defaults on a multilateral loses access to the institution that lends counter-cyclically, precisely when it most needs one.

For a project this matters directly. A development bank in the lending group lowers expected loss for everyone in the structure, which is why its participation changes pricing and tenor for the commercial lenders alongside it rather than only for its own tranche. It is also why the treatment is guarded so carefully by the institutions that benefit from it.

Blended finance, and the risk in it

Blended finance uses concessional public capital to make a transaction viable for commercial investors: a first-loss tranche, a guarantee, an interest subsidy, or grant funding for preparation. The intention is to move a project from unbankable to bankable and to bring in capital that would not otherwise arrive.

Two risks follow. The first is that the concession is larger than needed, so public money subsidises a return that would have been earned anyway. The second is adverse selection: the projects that need blending most are the ones commercial investors judged worst, and there is a reason for that judgement. The discipline is to ask what would have happened without the concessional element, which is the same additionality question the evaluation literature asks.

Block E discussion

Seminar questions

1. If a development institution lends at below the commercial rate, is it subsidising the project, the sponsor, or the commercial lenders alongside it?
2. Preferred creditor treatment is a convention rather than a legal right. What sustains it, and what would happen if it broke?
3. Project preparation is a public good for the market as a whole. Who should fund it, and does the answer differ between a middle income and a low income country?

Block F

Contracts that do not hold

Renegotiation, its causes, and what it does to the competition that preceded it.

Indicative length 30 minutes

The problem franchise bidding was supposed to solve

Competition for the market replaces competition in it: bidders compete at tender, and the winner is the one offering the lowest tariff or the highest payment. If the competition is genuine, the monopoly rent is competed away at the moment of award.

The argument holds only if the contract that is awarded is the contract that is performed. Where terms are renegotiated afterwards, the competition selected the bidder most willing to bid aggressively and renegotiate later, which is close to the opposite of what was intended.

How often concessions are renegotiated

Frequently

Guasch, Laffont and Straub analyse close to one thousand concessions awarded in Latin America and the Caribbean between 1989 and 2000, across telecommunications, energy, transport and water.

Renegotiation in their data is common rather than exceptional, and markedly more common in water and transport than in telecommunications and energy. It typically occurs within the first few years of a contract intended to run for decades.

Their model treats renegotiation as the consequence of imperfect enforcement, and tests what raises its probability. Regulatory institutions matter, contract design matters, and economic shocks matter, but so do features of the award itself.

The sector pattern is informative. Where technology changes quickly and competition is possible, contracts hold less well in theory and better in practice; where the asset is a genuine monopoly serving households, renegotiation is close to routine.

What raises the probability of renegotiation

- The absence of a regulator at the time of award, or a regulator without the independence to refuse.
- Award on the basis of the lowest tariff without qualification, which rewards aggressive bidding.
- Investment obligations specified in inputs rather than outputs, which invites argument about whether they were met.
- A price cap regime rather than rate of return, which places more risk on the concessionaire and therefore more pressure at the point of shock.
- Corruption and weak legal enforcement, both of which lower the cost of asking.
- Macroeconomic shocks, which supply the occasion even where they are not the cause.

Hold-up, and why sunk capital changes everything

Before the investment is made, the state faces a competitive market of potential concessionaires. After it is made, it faces one party whose capital cannot be moved. The relationship has been transformed from a competitive one into a bilateral monopoly, and both sides now have an incentive to reopen terms.

The state can lower the allowed tariff, knowing the asset will keep operating. The concessionaire can threaten to stop serving customers who vote. Neither threat is credible in isolation and both are used. Every device in a concession agreement, from termination compensation to step-in rights to dispute resolution, exists to manage the consequences of this transformation.

Designing against renegotiation

What helps

- An independent regulator in place before award, not created afterwards
- Award criteria that test deliverability as well as price
- Output specifications, which are harder to argue about than inputs
- Explicit and pre-agreed rules for the events most likely to trigger a request, including inflation, exchange rate movement and change in law
- Transparency, so that a renegotiation is publicly visible

What does not help

- Longer contracts, which increase the range of events not provided for
- More detailed specification, beyond the point where detail can be monitored
- Penalties that the state will not impose because the service must continue
- Assuming that arbitration resolves the matter. It resolves the legal question and not the political one
- Believing the tender price where it is materially below the rest of the field

Termination, and who pays for what

Cause	Typical compensation to the private party	Rationale
Authority default	Debt in full, equity return, sometimes lost profit	The state caused the loss and lenders will not fund otherwise
Contractor default	Debt only, often at a discount; equity takes the loss	Equity bore the performance risk and failed
Voluntary termination by the state	Debt in full and equity return, as for authority default	The state is buying its way out of a bargain it made
Force majeure	Debt, and equity contributions without a return	Neither side caused it, so the loss is shared
Corruption	Usually debt only, sometimes nothing	A deterrent, and often the hardest to enforce in practice

Reading a renegotiation

When a request arrives, four questions establish what is actually happening. What event is being cited, and was it allocated to a party in the contract. Is the request for more money, more time, a lower obligation, or a longer term, since the last of these is the easiest to grant and the most expensive. What is the alternative to agreeing, and is the state willing to face it. And who else is watching, because a concession granted once establishes what a request is worth in every other contract the state holds.

The fourth question is the one governments answer worst. Renegotiation is not a bilateral matter; it is a signal to a market.

Block F discussion

Seminar questions

1. If renegotiation is anticipated by everyone, what is the tender actually selecting for?
2. Would a shorter concession, say ten years rather than thirty, reduce renegotiation? What would it cost?
3. A concessionaire requests a tariff increase citing an exchange rate movement outside its control. On what basis should the state decide, and what should have been in the contract?

Block G

Heathrow and nationally significant infrastructure

How the United Kingdom consents a project too large for the ordinary planning system.

Indicative length 35 minutes

A separate regime for the largest projects

The Planning Act 2008 created a distinct consenting route for nationally significant infrastructure: airports above a threshold, major roads and railways, large generating stations, and similar. Rather than a local planning application, the promoter applies for a Development Consent Order, which is examined by an appointed panel and decided by the Secretary of State.

The design intent was to make decisions on projects of national importance faster and more predictable by separating the question of principle, settled in advance through a National Policy Statement, from the question of detail, settled in the examination. Whether it has achieved that is a live argument, and one the planning students will have met in PGR01.

The Development Consent Order process

Stage	What happens	Why it matters to an investor
Policy	A National Policy Statement establishes the need case and the principle	The question of whether to build is settled before the application, which removes the largest single risk
Pre-application	Statutory consultation with communities and statutory bodies	Long, expensive, and the stage where the scheme is actually shaped
Acceptance and examination	An Examining Authority runs a written and hearing-based process to a statutory timetable	A defined timetable is what makes the regime financeable at all
Recommendation and decision	The Examining Authority reports and the Secretary of State decides	A political decision at the end of a quasi-judicial process
Challenge	Judicial review of the decision or of the policy statement behind it	The residual risk. A consent is final only once the courts are finished

Heathrow expansion as a case in state and private partnership

Heathrow expansion is an unusually complete case for this week. The need case was settled through a national policy statement rather than a planning application. That statement was itself challenged in the courts, found unlawful at one level and reinstated at another, which is a demonstration that the policy stage carries litigation risk and not only the decision stage. The scheme is promoted by a privately owned airport operator whose charges are regulated by the Civil Aviation Authority, so the return on the investment depends on a regulatory determination that is separate from the consent. And the government's position has changed with governments.

Every element of this week is present in one project: policy, consent, regulation, litigation, and a state that is simultaneously the planning authority, the economic regulator, the policy-maker and, in effect, a commercial counterparty.

What a national policy statement settles, and what it does not

A national policy statement establishes the need for a category of infrastructure and the policy against which individual applications will be judged. Its purpose is to take the argument about whether the country needs, say, additional airport capacity out of the examination of a particular scheme, so that the examination can be about that scheme.

That is a genuine transfer of risk from the promoter to the state, and it is the main reason the regime is financeable. It is also why the policy statement, rather than the consent, has become the object of litigation: challenge the policy and every application resting on it falls with it.

What the litigation history teaches

- A policy statement can be quashed for a failure of process rather than for being substantively wrong, and the distinction matters to an investor only in that a process failure can usually be cured.
- Climate obligations have become a live ground of challenge for high-emitting infrastructure, which converts a policy commitment into a legal constraint on consenting.
- Litigation adds years, and years at the pre-construction stage are expensive because the equity is committed and earning nothing.
- A promoter cannot insure against this, and cannot easily price it, which is why policy risk is the risk most often cited by investors as the reason they will not fund greenfield infrastructure.
- The remedy, if there is one, is a policy framework robust enough to survive challenge rather than a faster court.

Surface access, and the coordination problem

A major airport expansion depends on road and rail capacity that the promoter does not control and cannot build. That capacity is funded by a different department, appraised under a different business case, and delivered on a different timetable, and its benefits are shared with users who have nothing to do with the airport.

This is the coordination institution in PIMA, and its failure is one of the most common reasons infrastructure underdelivers. A scheme appraised on the assumption that complementary investment arrives, where nothing obliges anyone to provide it, has embedded an assumption it cannot enforce.

Where the state meets the private partner

As policy-maker

Setting the national policy statement that establishes whether the project should happen at all, and revising it when policy changes.

As consenting authority

Running the examination and taking the decision, through a process that must be, and must be seen to be, procedurally fair.

As economic regulator

Determining the charges the operator may levy, and therefore whether the investment earns a return, through a separate process on a separate timetable.

As guarantor of surface access

Delivering the road and rail capacity the scheme depends on, which is a different department with its own budget and its own appraisal.

As climate policy-maker

Setting the carbon budgets against which the scheme's emissions are assessed, which is what made the policy statement litigable.

As a party with an interest

Wanting the investment to happen for growth reasons, while running the processes that decide whether it may.

Questions to put to the guest

- Where in the process does a scheme of this kind actually get decided, as opposed to where the statute says it is decided?
- How does a promoter manage a consenting process that outlasts several governments and their policies?
- What can be committed to a private partner before consent is granted, and what has to wait?
- How is the interaction between the consent timetable and the regulatory determination managed, given that neither can wait for the other?
- What would you change about the regime, and what is good about it that outsiders miss?

Block G discussion

Seminar questions

1. The Development Consent Order regime settles the principle in advance through a policy statement. What risk does that transfer, from whom to whom?
2. Heathrow expansion has been supported, refused, litigated and revived. What would a rational investor have done at each turn, and what does that imply about the cost of policy instability?
3. The state appears in this case as policy-maker, consenting authority, regulator, infrastructure provider and interested party. Is that a problem to be solved or a fact to be managed?

Block H

Reserve material

Slides held back from the main sequence.

Indicative length As required

Candidate cases for Week 2

Case	What it illustrates	Availability of material
Heathrow expansion	Consenting, policy risk and litigation	Extensive, and a guest speaker
England and Wales water	Price control mechanics and regulatory hostility	Ofwat determinations and appeals are public
Norwegian quality assurance	A gateway that precedes political commitment	Flyvbjerg ch. 18 and Ministry of Finance material
Chilean concessions programme	Franchise bidding at scale, and renegotiation	Guasch and successor literature
Thames Tideway	A bespoke regulatory and financing structure for one asset	Public documentation is unusually complete
UK offshore transmission	Regulated revenue created by competitive tender	Ofgem tender documents and evaluations

An alternative running order

Where the cohort is drawn mainly from Real Estate Finance, block C can be expanded and taught first, with the price control arithmetic on the board, and blocks A and B compressed into a single thirty-minute framing. The institutional material then reappears in block G, where the Heathrow case makes it concrete rather than abstract.

Where the cohort is mainly from Planning or Environmental Policy, keep A and B intact, compress C to the building blocks table and the allowed return slide, and give the recovered time to F, since renegotiation is the part of the week least covered elsewhere in the MPhil.

Sources drawn on for this deck

- Delmon, J., Private Sector Investment in Infrastructure, chapters 1, 4, 5, 6, 7, 8, 10 and 15.
- IMF, Public Investment Management Assessment Handbook, chapters 1 and 2.
- APMG PPP Certification Program Guide, chapters 2 to 5.
- Flyvbjerg, B. (ed.) (2017), The Oxford Handbook of Megaproject Management, chapter 18.
- Abor, J. Y. and others (eds) (2025), The Routledge Handbook of Infrastructure Finance, chapters 16, 29, 30 and 32.
- Journal articles as cited on each slide. Full references in the module reading list.

Where Week 2 leaves us

The case for state involvement rests on four market failures, and none of them says whether the state should own, regulate or contract.

Selection is where the value is made or lost, and the institutions that select are the subject rather than the background.

A price control is a weighted average cost of capital set administratively, which converts market risk into political risk rather than removing it.

The competition that precedes a concession is undone by the renegotiation that follows it, unless the institutions to prevent that were built first.